

The background is a deep blue, textured surface. It features several large, stylized neurons with multiple branching processes. Interspersed among the neurons are several atomic models, each with a central orange sphere and three elliptical orbits containing smaller blue and orange spheres. In the background, there are faint, dark silhouettes of human figures in various running or walking poses.

Neural Network Potentials

under the hood

Literature Meeting

Oct 2025

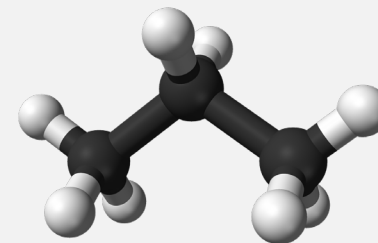
We have enough physics to predict biology and chemistry, but not enough time



Courtesy of Monica Dirac

Former Member Paul Dirac, pictured here on the grounds of the Institute around 1962, began his first year-long sabbatical at the Institute in 1934. The youngest-ever theoretician to win the Nobel Prize in Physics, Dirac would often set off on Saturday mornings with an axe over his shoulder to help Oswald Veblen, one of the Institute's first Faculty members, and others clear paths in the Institute Woods.

The underlying physical laws necessary for the mathematical theory of a large part of physics and the whole of chemistry are thus completely known, and the difficulty is only that the exact application of these laws leads to equations that are much too complicated to be soluble. It therefore becomes desirable that approximate practical methods of applying quantum mechanics should be developed, which can lead to an explanation of the main features of complex atomic systems without too much computation.



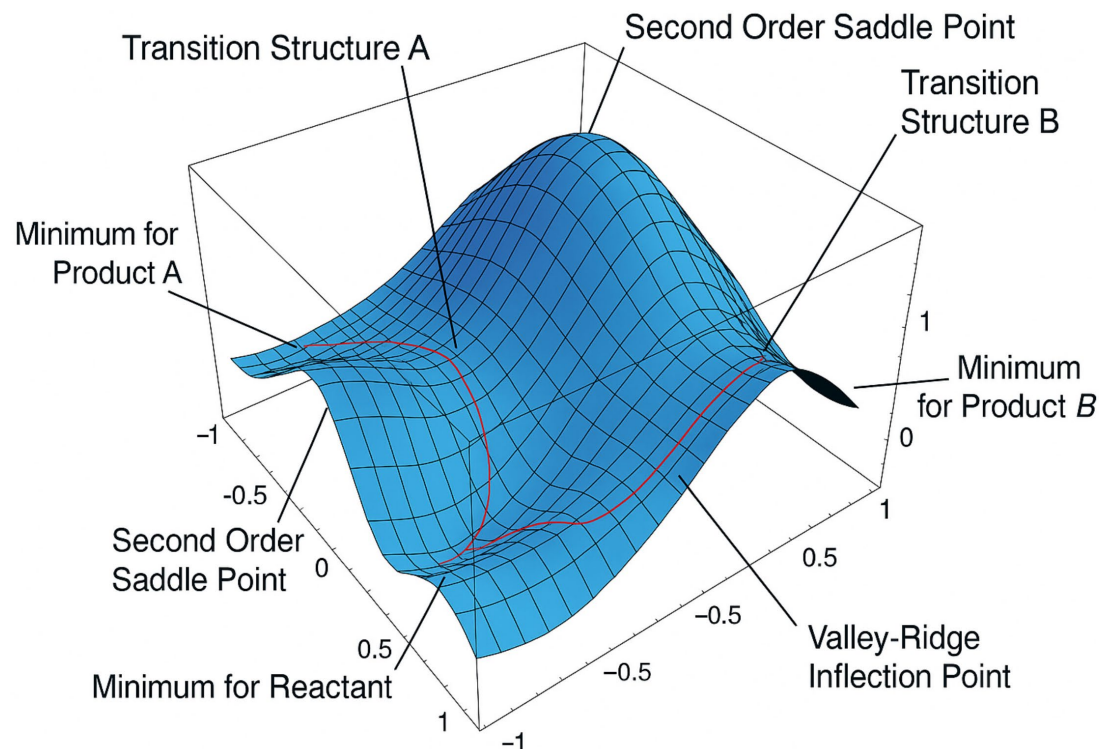
Full configuration interaction (CI) for C₃H₈:

1.3 trillion determinants

6.6 years with B–O approximation

The key goal in computational chemistry is to model potential energy surfaces

$$E = f(\mathbf{r}, \mathbf{R})$$



Ultimately, we want to predict physical quantities

Energy

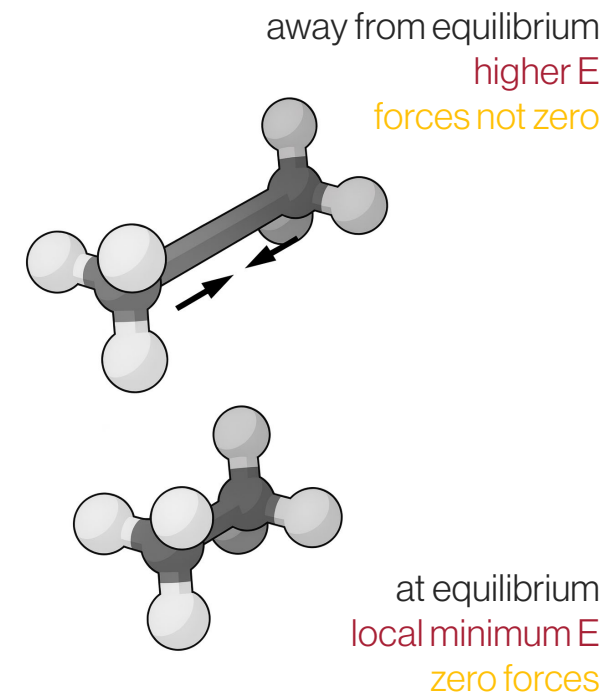
$$H_{\text{tot}}(\mathbf{r}, \mathbf{R}) \Psi_{\text{tot}}(\mathbf{r}, \mathbf{R}) = E_{\text{tot}} \Psi_{\text{tot}}(\mathbf{r}, \mathbf{R})$$

Forces

$$\mathbf{F} = -\nabla E = -\left(\frac{\partial E}{\partial x}\mathbf{i} + \frac{\partial E}{\partial y}\mathbf{j} + \frac{\partial E}{\partial z}\mathbf{k}\right)$$

Others

Dipole, charge, etc.



Tasks

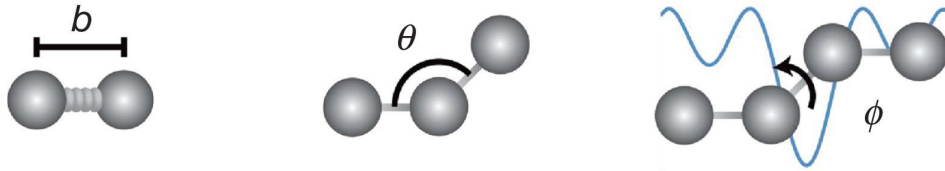
1. Identify stationary points (zero forces/gradient in all directions)
2. Characterize stationary points (second-derivative matrix, Hessian)
3. Minima (stable geometries) versus saddle points (transition states)
4. Calculate relative energies of minima and reaction barriers

Traditional approaches involve force fields or quantum chemistry

Force fields

Energies described by simple Newtonian/Coulombic potentials

$$V(R)_{\text{bonded}} = \sum_{\text{bonded}} K_b (b - b_0)^2 + \sum_{\text{angles}} K_\theta (\theta - \theta_0)^2 + \sum_{\text{torsions}} \sum_n k_\phi \cos(n\phi - \gamma)$$



$$V(R)_{\text{nonbonded}} = \sum_{\text{van der waals}} \left(\frac{A}{r_{ij}^{12}} - \frac{B}{r_{ij}^6} \right) + \sum_{\text{electrostatic}} \left(\frac{q_i q_j}{\epsilon_D r_{ij}} \right)$$



Limitations

Low “expressivity”: a small number of terms cannot capture nuances

Not general

Cannot capture electronic redistribution (bond changes)

Electronic structure theory (specifically DFT)

Ground-state electron density maps one-to-one with wavefunction/energy

$$E[\rho(\mathbf{r})] = \sum_i^N \left(\left\langle \chi_i \left| -\frac{1}{2} \nabla^2 \right| \chi_i \right\rangle - \sum_A^M \left\langle \chi_i \left| \frac{Z_A}{r_{iA}} \right| \chi_i \right\rangle \right) + \sum_i^N \left\langle \chi_i \left| \frac{1}{2} \int \frac{\rho(\mathbf{r}') d\mathbf{r}'}{|\mathbf{r}_i - \mathbf{r}'|} \right| \chi_i \right\rangle + E_{\text{xc}}[\rho(\mathbf{r})]$$

non-interacting
kinetic energy

nuclear-electron
attraction

classical Coulomb
electron-electron
repulsion

exchange-correlation
energy functional

$$E_{\text{xc}}^{\text{B3LYP}} = (1 - a) E_{\text{x}}^{\text{LSDA}} + a E_{\text{x}}^{\text{HF}} + b \Delta E_{\text{x}}^{\text{B}} + (1 - c) E_{\text{c}}^{\text{LSDA}} + c E_{\text{c}}^{\text{LYP}}$$

Limitations

Slow

Empirically/Serendipitously designed functionals

Traditional approaches involve force fields or quantum chemistry

Force fields

Energies described by simple Newtonian/Coulombic potentials

$$V(R)_{\text{bonded}} = \sum_{\text{bonded}} K_b (b - b_0)^2 + \sum_{\text{angles}} K_\theta (\theta - \theta_0)^2 + \sum_{\text{torsions}} \sum_n k_\phi \cos(n\phi - \gamma)$$



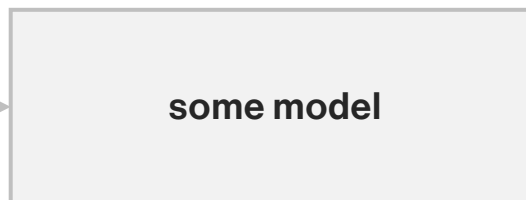
Machine learning?

Using lots of data and sophisticated models, aim to capture “rules of physics” pertaining to PESs of atomic systems

$$V(R)_{\text{nonbonded}} = \sum_{\text{van der waals}} \left(\frac{A}{r_{ij}^{12}} - \frac{B}{r_{ij}^6} \right) + \sum_{\text{electrostatic}} \left(\frac{q_i q_j}{\epsilon_D r_{ij}} \right)$$

High expressivity

Slow to train, fast to infer



Limitations

Low “expressivity”: a small number of terms cannot capture nuances

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Electronic structure theory (specifically DFT)

Ground-state electron density maps one-to-one with wavefunction/energy

$$E[\rho(\mathbf{r})] = \sum_i \left(\left\langle \chi_i \left| -\frac{1}{2} \nabla^2 \right| \chi_i \right\rangle - \sum_A \left\langle \chi_i \left| \frac{Z_A}{r_{iA}} \right| \chi_i \right\rangle \right) + \sum_i \left\langle \chi_i \left| \frac{1}{2} \int \frac{\rho(\mathbf{r}') d\mathbf{r}'}{|\mathbf{r}_i - \mathbf{r}'|} \right| \chi_i \right\rangle + E_{xc}[\rho(\mathbf{r})]$$

non-interacting
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$$E_{xc}^{\text{B3LYP}} = (1-a)E_{xc}^{\text{LSDA}} + aE_x^{\text{HF}} + b\Delta E_x^{\text{B}} + cE_c^{\text{LYP}}$$

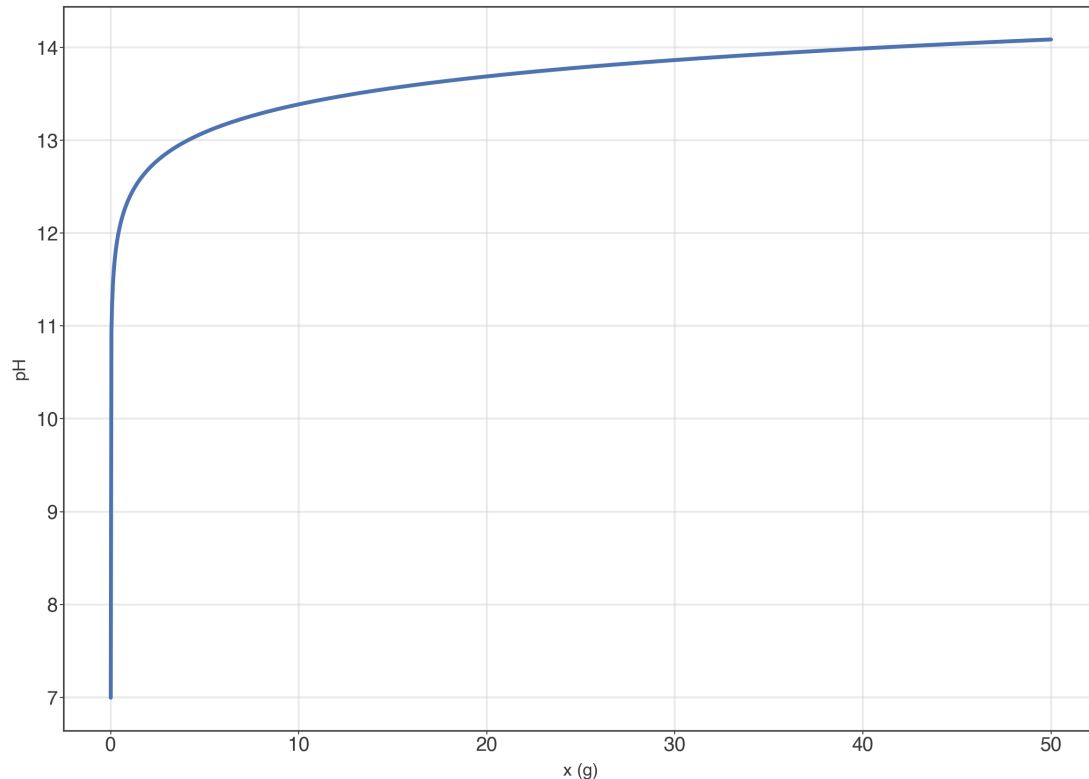
Limitations

Slow

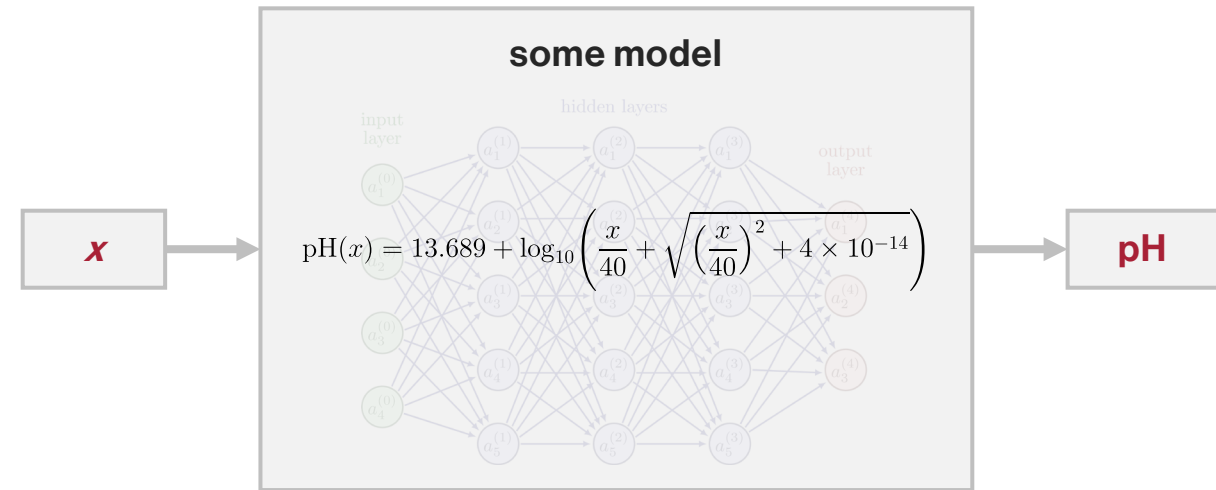
Empirically/Serendipitously designed functionals

Physical phenomena can be modeled empirically

If we dissolve **x** grams of NaOH to make a 1 L aqueous solution at room temperature, what is the **pH** of the solution?



A regression/NN model can learn the function knowing nothing about the physical basis!



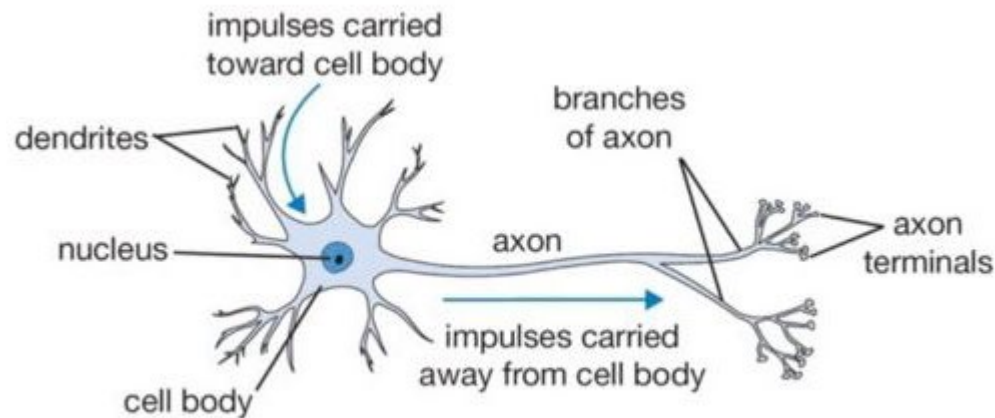
Fails if we change the temperature/solvent, or if we go above the solubility of NaOH in water.

But if we could collect enough data on all the variations we possibly care about, we could end up doing pretty well!

Neural networks are simply many-layered, nonlinear regression models

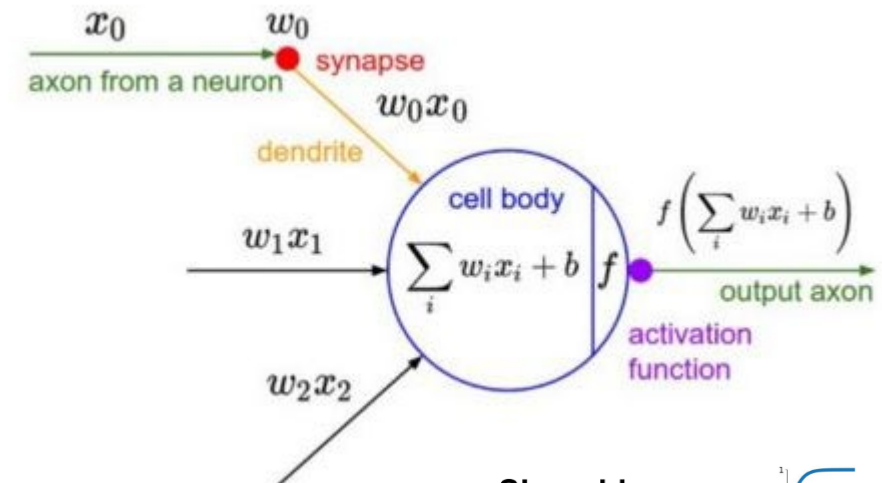
Inspiration from biology

Neurons sum up multiple inputs
If the sum exceeds the activation threshold, they fire.

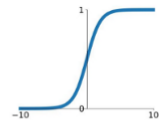


Artificial neuron

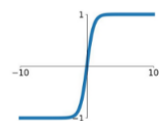
Sums up multiple inputs x with associated weights w
If the sum exceeds the activation threshold, the output is nonzero.



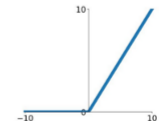
Sigmoid
 $\sigma(x) = \frac{1}{1+e^{-x}}$



tanh
 $\tanh(x)$



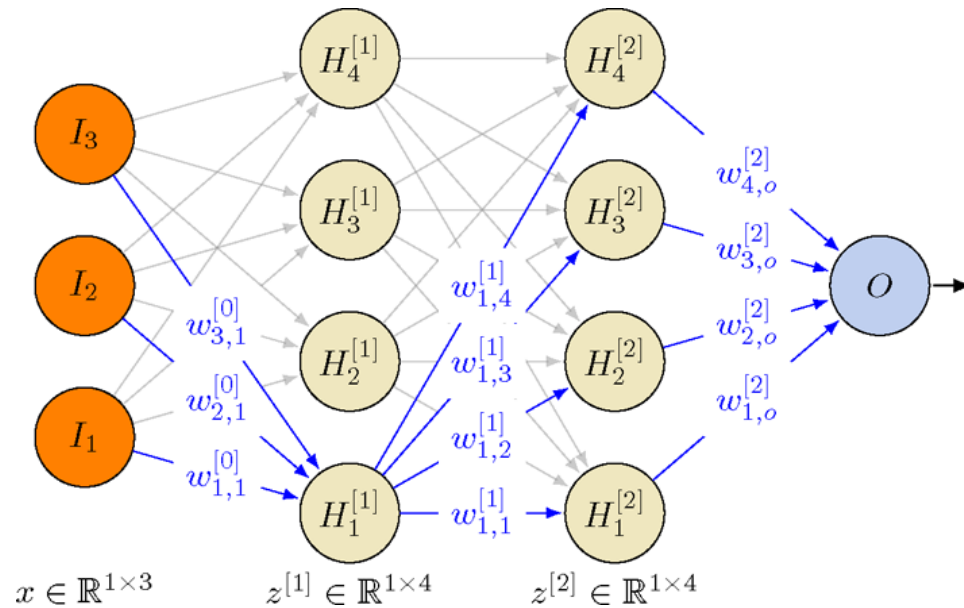
ReLU
 $\max(0, x)$



NNs are simply many-layered, nonlinear regression models

A simple multilayer feed-forward NN

Each weight w is a “parameter”



GPT5 has 10^{11-12} parameters

Training

Given lots of (input, output) pairs, find the right set of parameters $\mathbf{w}$ that maximizes performance (model output is close to correct output)

A multidimensional optimization problem

Model	Training end	Chip type	TFLOP/s (max)	Chip count	Wall clock (days)	Total time (years)	Retail (US\$)	MMLU
GPT-3 175B	Apr/2020	V100	130	10,000	15 days	405y	\$9M	43.9
Llama 1 65B	Jan/2023	A100	312	2,048	21 days	118y	\$4M	63.4
Llama 2 70B	Jun/2023	A100	312	2,048	35 days	196y	\$7M	68.0
Titan 200B	Apr/2023	A100	312	13,760	48 days	1,319y	\$45M	70.4
GPT-4 1.7T	Aug/2022	A100	312	25,000	95 days	6,507y	\$224M	86.4
Gemini	Nov/2023	TPUv4	275	57,000	100 days	15,616y	\$440M	90.0
Llama 3 405B	Apr/2024	H100	989	24,576	50 days	3,366y	\$125M	85+
GPT-5	Apr/2024	H100	989	50,000	120 days	16,438y	\$612M	
Grok 2	Jun/2024	H100	989	20,000	50 days	6,571y	\$245M	
Olympus	Aug/2024	H100	989					
Gemini 2	Nov/2024	TPUv6	1,847					
Grok 3	Dec/2024	H100	989	100,000	50 days	32,855y	\$1.2B	

Alan D. Thompson, May/2024, LifeArchitect.ai

Table. Model training compute (see working, with sources⁸).

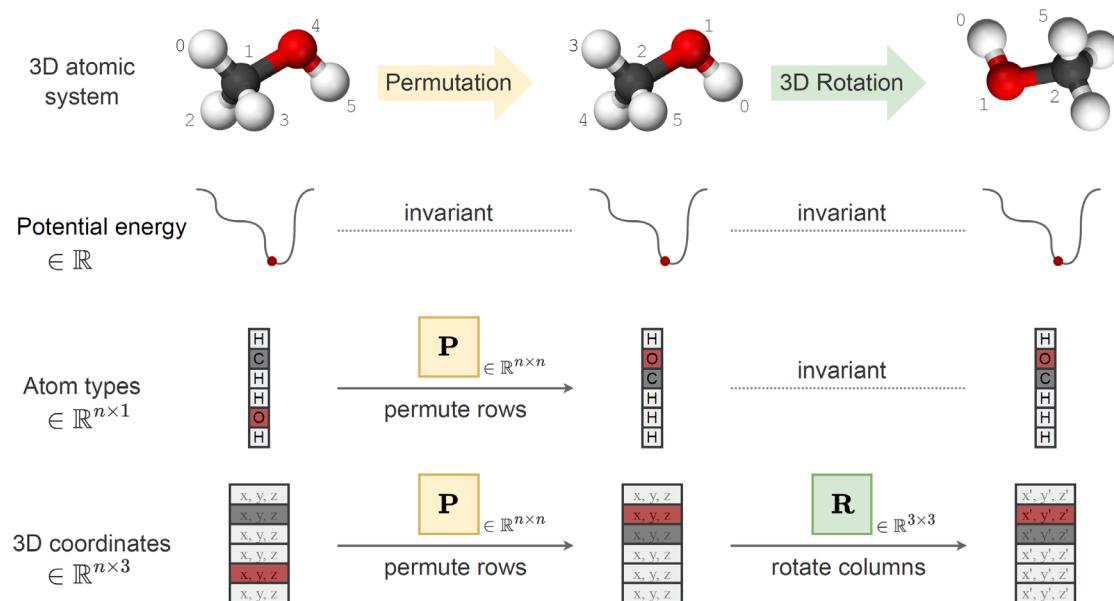
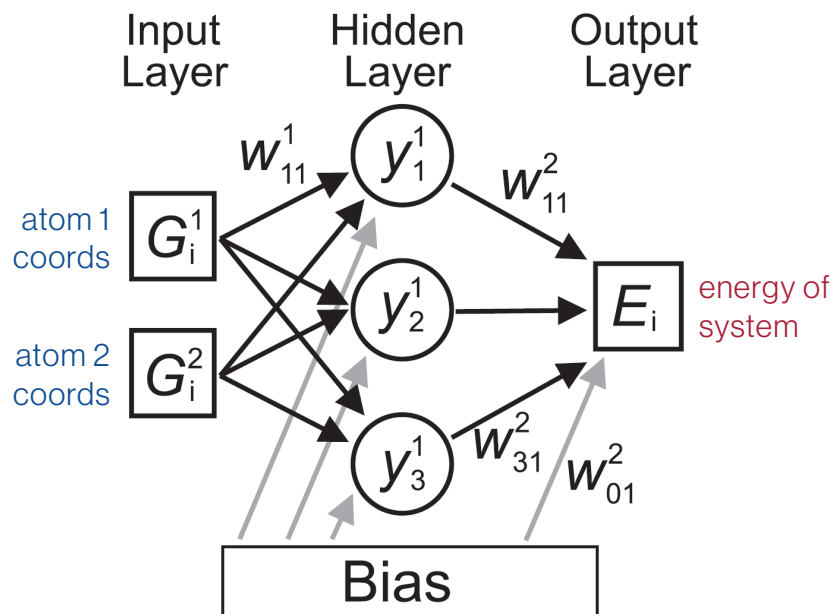
Inference

Given an input, pass it through the trained NN with optimized parameters, collect an output

Very fast even for large models (e.g. GPT response time)

NNs can be designed to represent chemical PESs

A simple NN is unsuitable for PESs

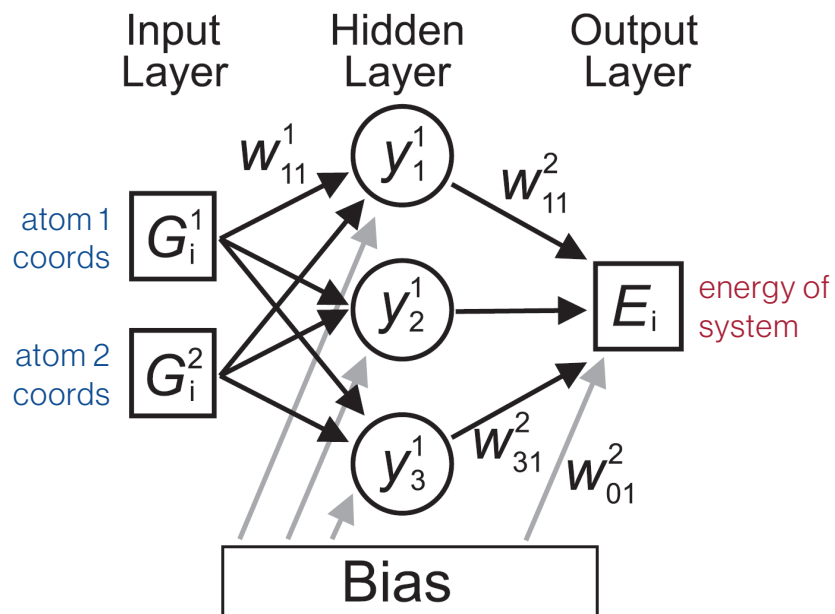


Since all weights are different, order of atom 1/atom 2 is not arbitrary, so changing order of input changes energy

Fixed network structure: NN optimized for 2-atom PES cannot be used for a different system size

NNs can be designed to represent chemical PESs

A simple NN is unsuitable for PESs

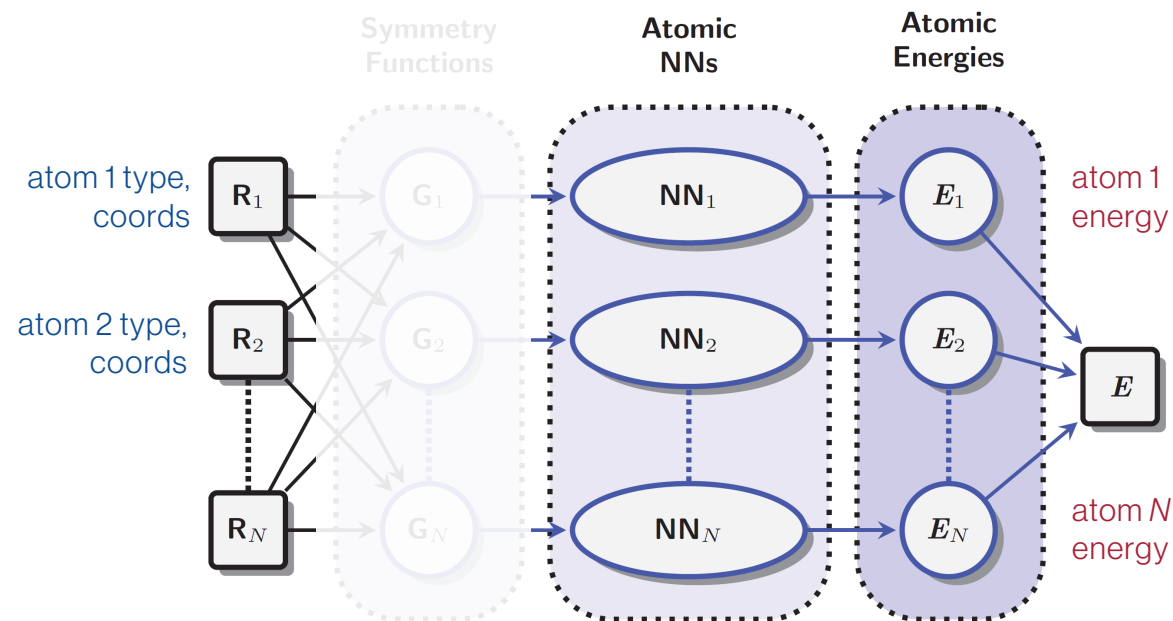


Since all weights are different, order of atom 1/atom 2 is not arbitrary, so changing order of input changes energy

Fixed network structure: NN optimized for 2-atom PES cannot be used for a different system size

Key innovation 1:

Energy is the sum of atomic energies predicted from atomic NNs



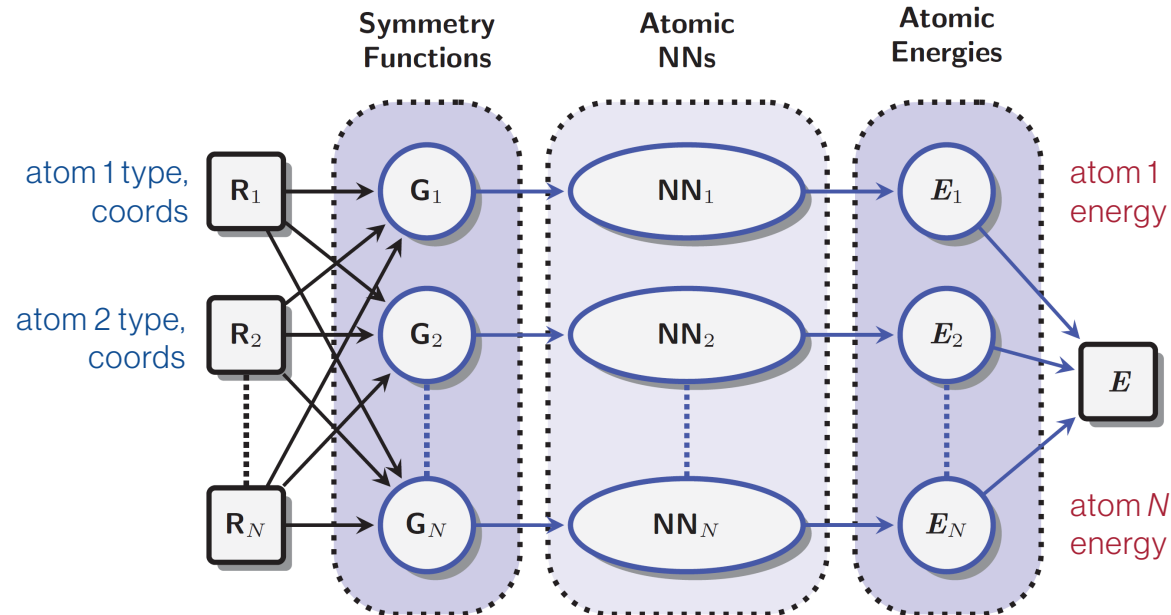
Every atom of the same element has the same atomic NN: *invariant* wrt atom permutations

Can be extended to any number of arbitrary atoms

Defining an energetically relevant local environment is key

Key innovation 2:

Symmetry function describes atom's local environment

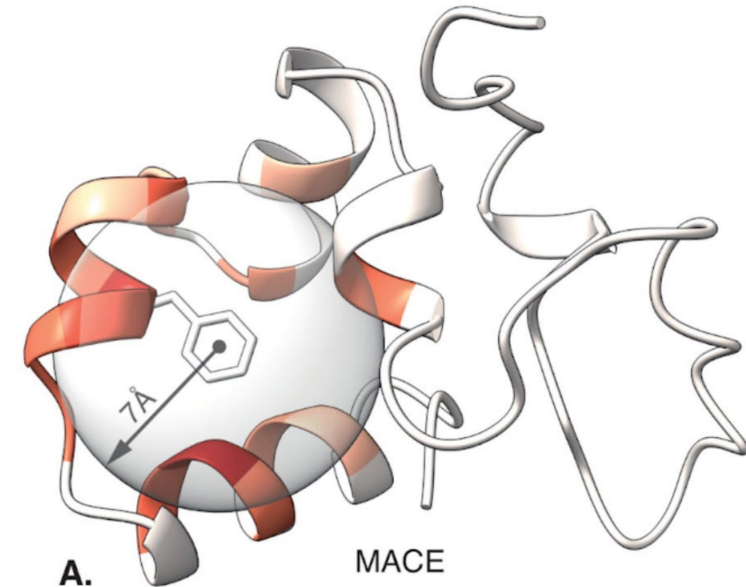


Define a cutoff radius beyond which atoms are invisible

$$f_c(R_{ij}) = \begin{cases} 0.5 \times \left[\cos\left(\frac{\pi R_{ij}}{R_c}\right) + 1 \right] & \text{for } R_{ij} \leq R_c, \\ 0 & \text{for } R_{ij} > R_c. \end{cases}$$

Within local environment (version 1):

Radial functions describe distances between pairs
Angular functions describe angles between all triples



By accounting for only local environment and with the right functional form, symmetry functions confer *invariance* wrt rotation/translation of the system

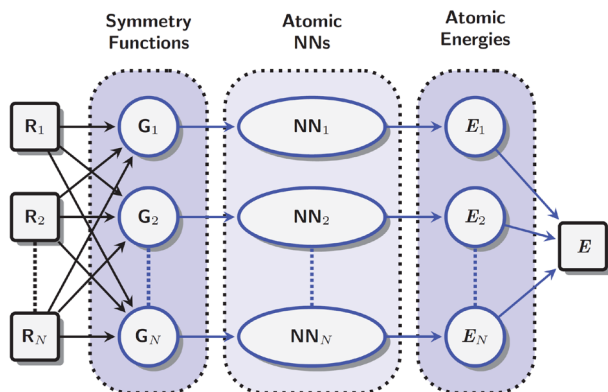
Does not account for long-range interactions

Works rather well on early examples (liquid phase Si, water dimer)

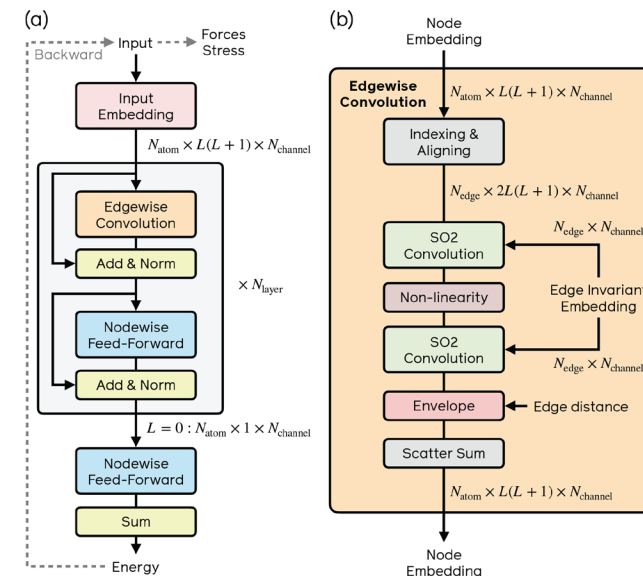
Note: this NN outputs energy; forces can be obtained by differentiating backwards through the NN (aka *backpropagation*)

Modern NNPs build on the Behler–Parrinello philosophy

2007



2025



Key research/advance areas

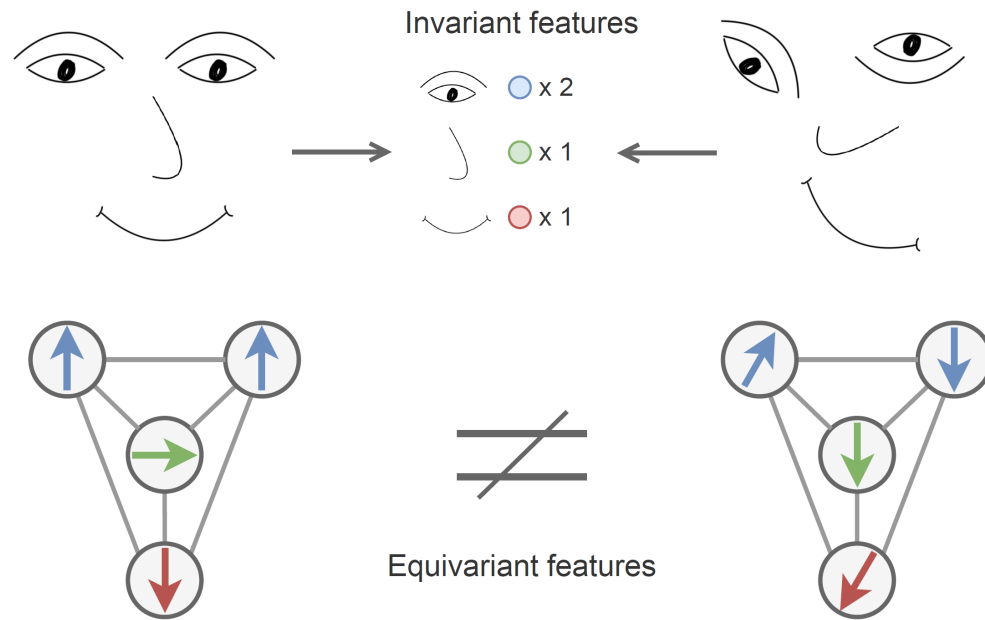
- 1 Input representation
- 2 Architecture
- 3 Energy conservation
- 4 Data
- 5 Long-range interactions

Remarkably, the core philosophy remains the same!

Many design choices to balance expressivity, cost, and accuracy.

Invariance is not enough; equivariance is key (to all modern ML!)

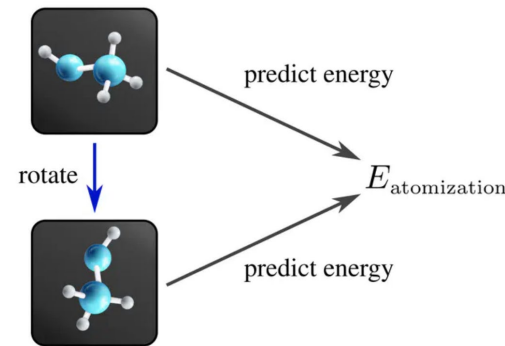
1 Input representation



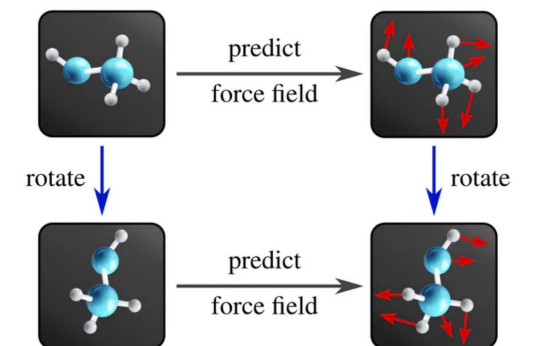
The Picasso Problem

Identifying a face requires understanding the *relative orientation* of the parts (equivariant information), not just detecting their presence (invariant features)

Energy is *invariant* to rotation



Forces are *equivariant* to rotation



Considerations and design principles

Each quantity in each hidden layer is associated with a type (invariant/equivariant)

Different types of quantities cannot be added together

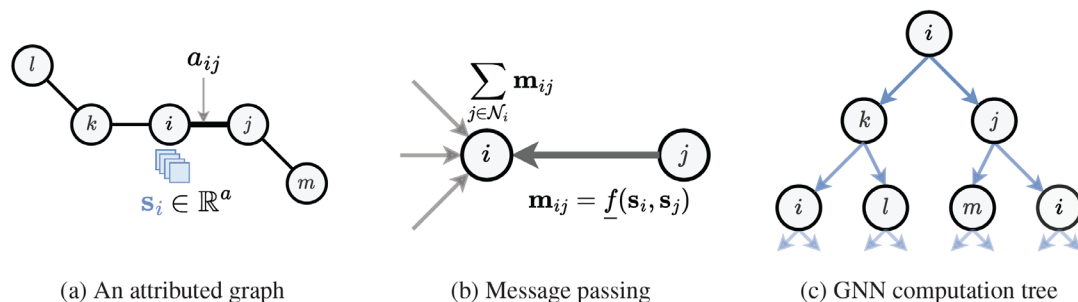
Special multiplication rules apply to different types (tensor product)

Cannot apply nonlinear operations (e.g. activation functions)

Geometric graph neural networks (GNNs) underlie modern NNPs

1 Input representation 2 Architecture

Graph neural networks (GNNs)



Each node contains a set of features $\mathbf{s}$

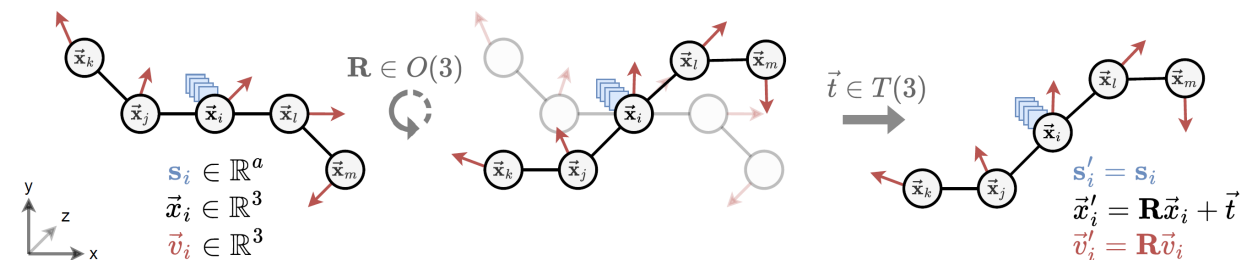
In each message-passing round, each node does:

1. **Message:** computes a message for each neighbor based on its own $\mathbf{s}$, neighbor's $\mathbf{s}$, and the edge connecting them
2. **Aggregate:** compiles incoming messages
3. **Update:** updates its own $\mathbf{s}$

Repeat (more rounds means a larger radius of communication)

GNNs are naturally important for network-based applications, e.g. social networks, protein folding (AlphaFold)

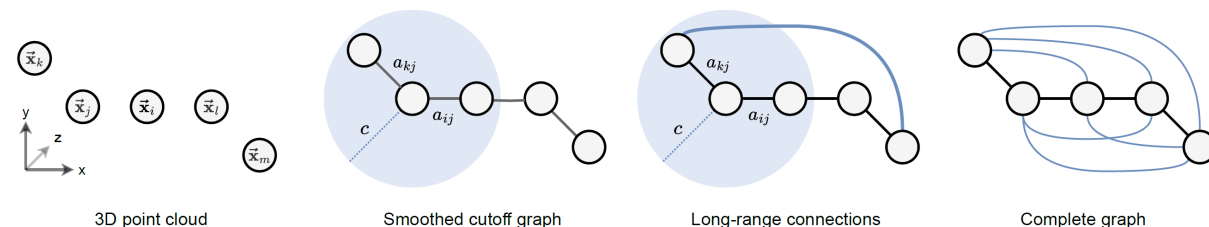
Geometric GNNs



In addition to scalar features $\mathbf{s}$ (simple quantities), each node also contains geometric attributes like coordinates $\mathbf{x}$ and other vector quantities (e.g. forces)

Apply invariance/equivariance considerations (see last slide);

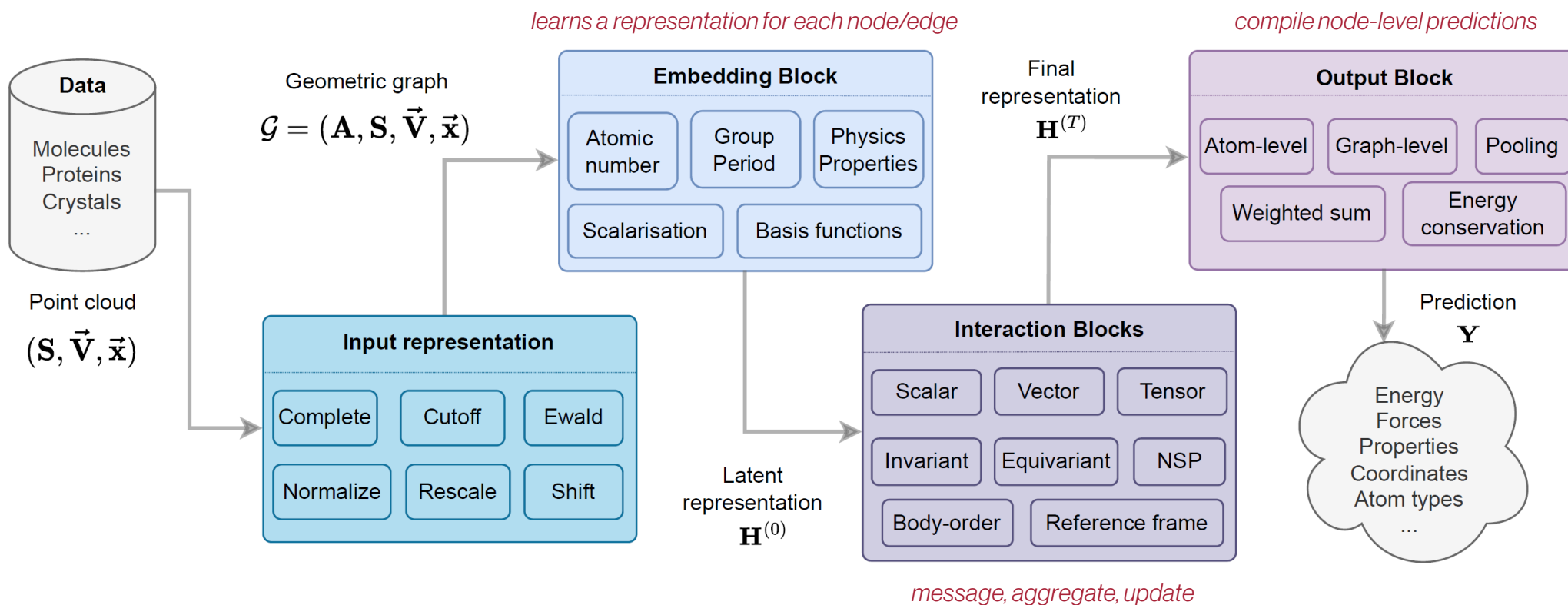
When should two nodes be connected by an edge?



Modern architectures *learn* the best atom representations during training

1 Input representation 2 Architecture

B-P represents atoms as symmetry functions; modern NNPs instead *learn representations*.
Meaningful atom representations are important for good performance, in addition to good parameters/architecture



Should the model predict energies or forces?

3 Energy conservation

Forces are important for geometry optimizations, molecular dynamics, etc.

Option 1: Predict energy

Using the B-P framework, forces can be obtained by differentiating back through the network, but this is costly (e.g. 2–4x memory)

Guarantees energy conservation:

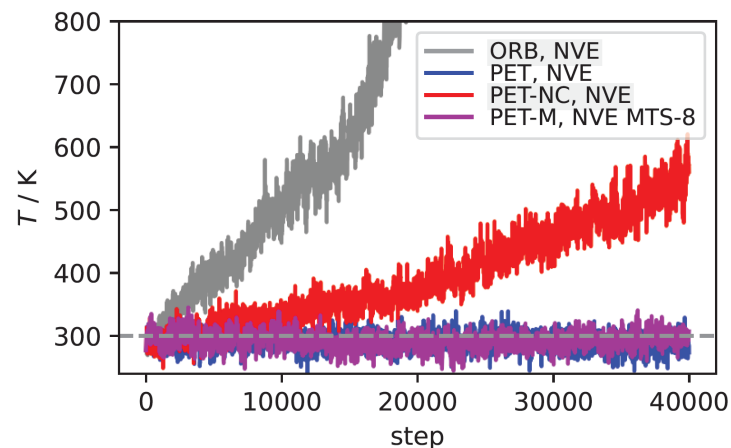
$$W = \oint \sum_{j=1}^N \mathbf{f}_j \cdot d\mathbf{r}_j = 0$$

Option 2: Predict forces

Avoids backprop, greatly accelerates training

Most optimizations rely on forces going to zero, but this is not guaranteed

Energies may run away during MD



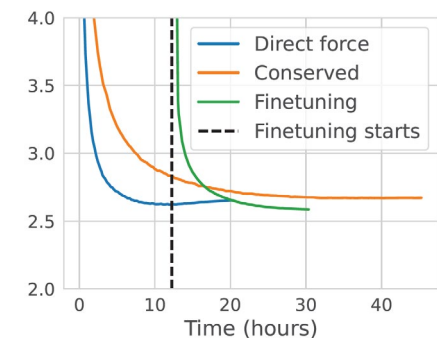
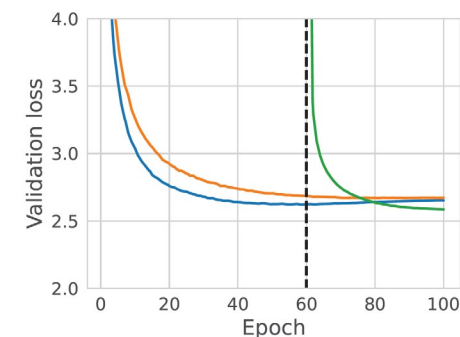
T rises 7 trillion K/sec with unconstrained model

Figure 2. Time series for the kinetic temperature along a NVE MD trajectory, for ORB, the conservative and non-conservative PET models, and for a PET-M model using a multiple time-stepping (MTS) algorithm that evaluates conservative forces every 8 steps.

Option 3: Direct-force pretraining

Two-phase training (implemented in eSEN):

1. Direct-force model
2. Fine-tune with conservative force prediction

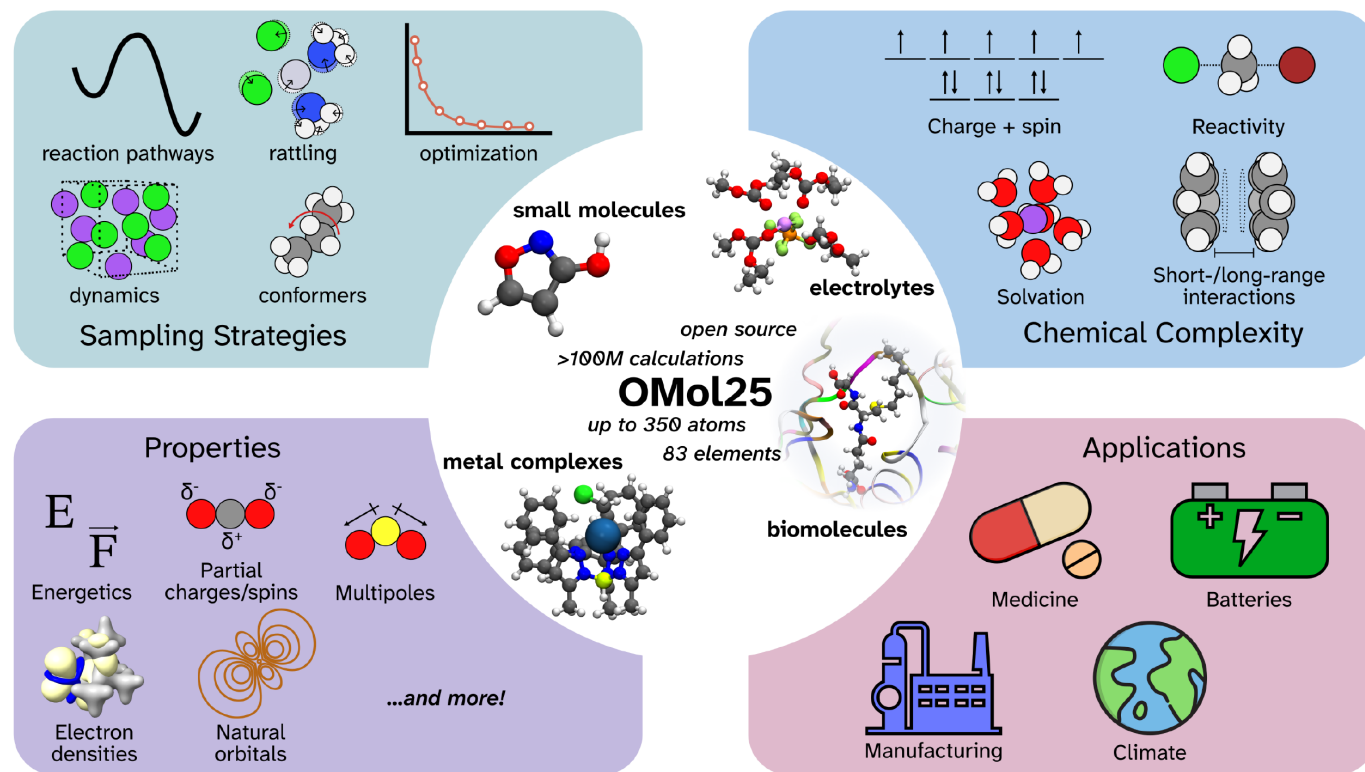


Significant expansion in training data enables better models

4 Data

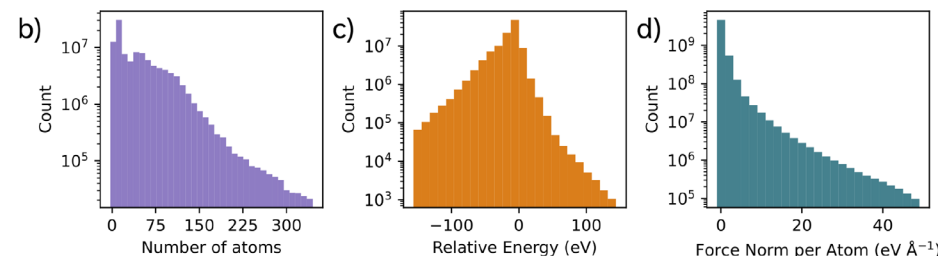
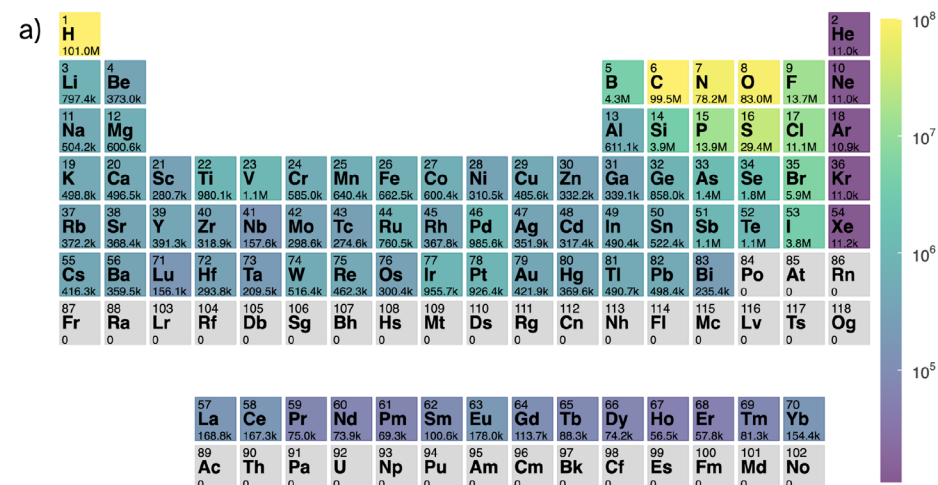
B-P and other earlier models were trained on a single type of system

“Foundation models” aim to learn agnostically across a broad scope of systems and tasks



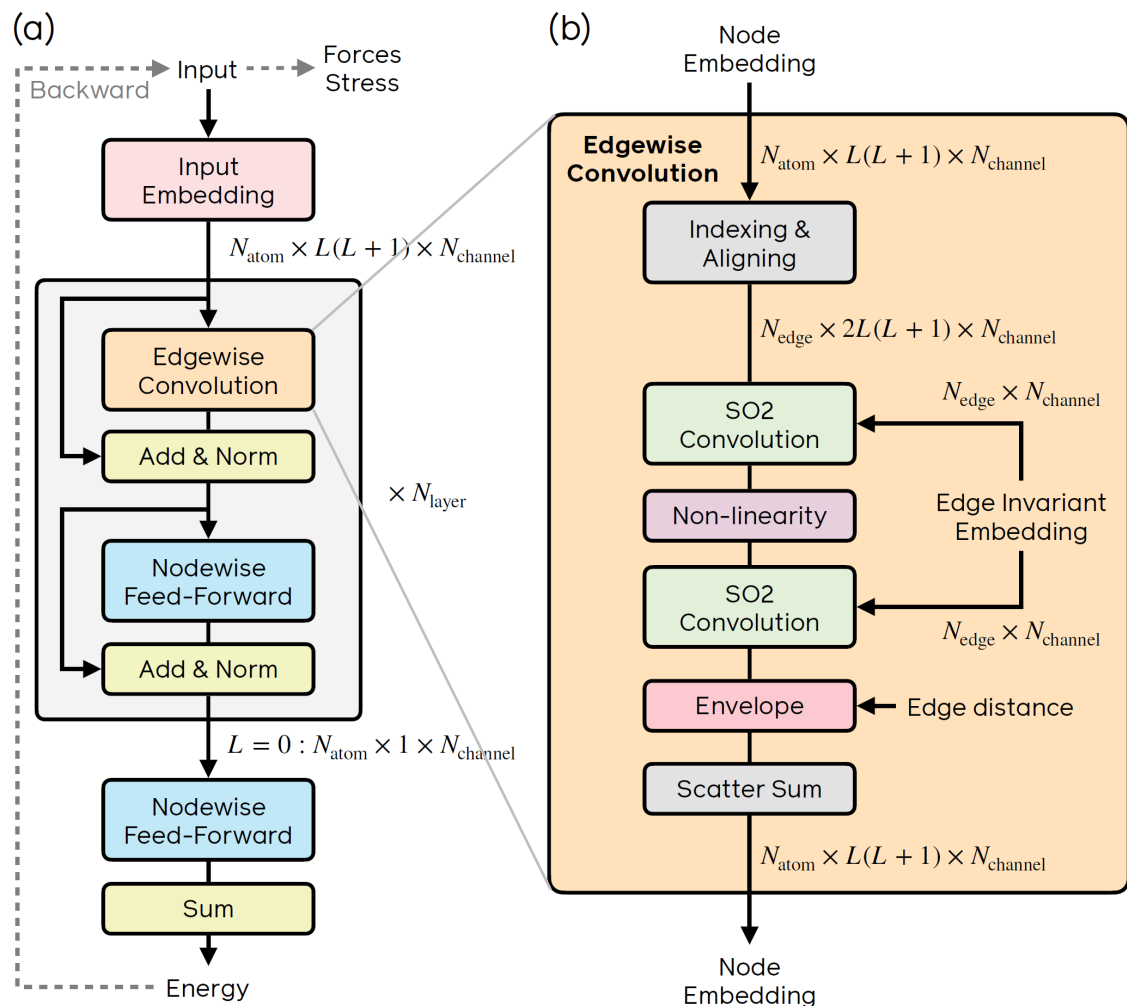
100 million DFT single-points at ω B97M-V/def2-TZVPD
(~10x previous efforts)

varying charge and spin states, explicit solvation,
reactivity, and various intermolecular interactions

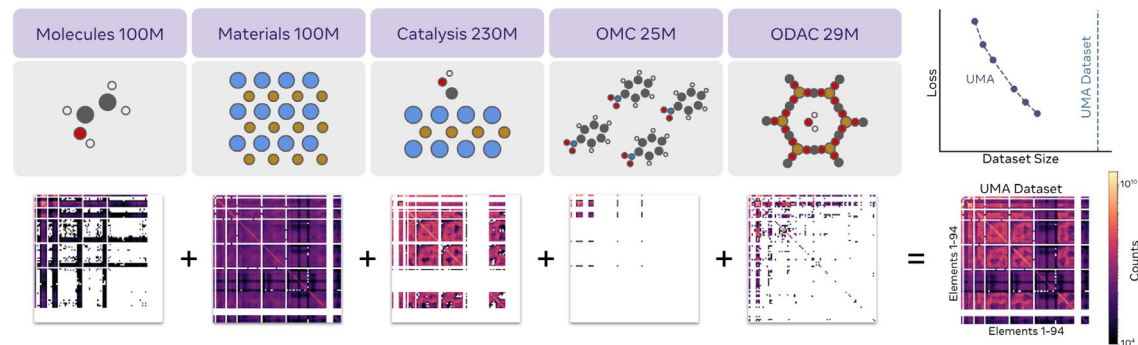


FAIR at Meta's Universal Model for Atoms (UMA) is a big step forward

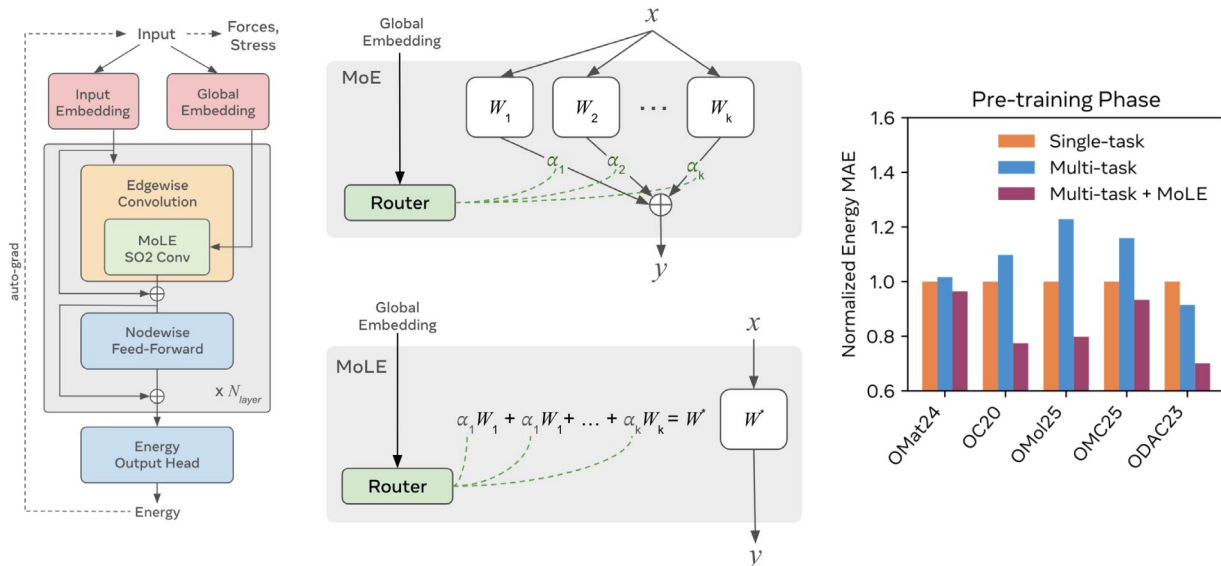
eSEN architecture



~500 million atomic systems



Mixture of Linear Experts: train five models for respective tasks that communicate with each other



UMA outperforms routine DFT methods on benchmarks

GMTKN55 (exhaustive properties, reaction energies, NCIs)

Name	Small Systems ↓	Large Systems ↓	Barrier Heights ↓	Intramolecular NCIs ↓	Intermolecular NCIs ↓	All ↓
<u>ωB97M-D3BJ/def2-QZVP</u>	2.13	5.80	2.40	4.72	4.30	4.10
<u>OMol25's eSEN Conserving Small</u>	7.20	4.81	2.93	5.20	6.93	5.58
<u>UMA Medium 1.1 (OMol task)</u>	8.08	4.27	2.91	4.92	7.92	5.79
<u>B3LYP-D3BJ/def2-QZVP</u>	5.03	11.45	3.65	5.68	5.56	6.20
<u>UMA Small 1.1 (OMol task)</u>	5.83	4.10	2.78	5.10	12.64	6.69
<u>AIQM2</u>	6.22	4.02	5.37	9.25	6.55	6.78
<u>UMA Small (OMol task)</u>	5.44	3.83	2.84	5.23	15.13	7.28
<u>AIQM1</u>	3.06	6.62	11.82	10.42	5.78	8.21
<u>OrbMol (Orb-v3 Conservative OMol)</u>	10.11	3.95	2.94	5.19	14.58	8.26
<u>MACE-ANI-CC</u>		7.67		13.78	3.02	9.28

Folmsbee (relative conformer energies)

Name	Overall MAE ↓	Overall RMSE ↓	Mean R ² ↑
<u>UMA Medium 1.1 (OMol task)</u>	0.21	0.33	0.85
<u>OMol25's eSEN Conserving Small</u>	0.21	0.33	0.85
<u>UMA Small 1.1 (OMol task)</u>	0.21	0.34	0.85
<u>UMA Small (OMol task)</u>	0.21	0.34	0.85
<u>OrbMol (Orb-v3 Conservative OMol)</u>	0.22	0.35	0.83
<u>ωB97X-D/def2-TZVP</u>	0.24	0.39	0.83
<u>B3LYP/def2-TZVP</u>	0.25	0.41	0.83
<u>B97-3c</u>	0.30	0.53	0.80
<u>Egret-1e</u>	0.31	0.50	0.78
<u>Egret-1</u>	0.31	0.53	0.78
<u>MACE-ANI-CC</u>	0.32	0.55	0.77

Wiggle150 (highly strained conformers)

Name	Overall MAE ↓	Overall RMSE ↓
<u>ωB97M-V/def2-QZVP</u>	0.87	1.18
<u>UMA Medium 1.1 (OMol task)</u>	0.88	1.21
<u>UMA Small 1.1 (OMol task)</u>	0.90	1.22
<u>UMA Small (OMol task)</u>	0.91	1.23
<u>OMol25's eSEN Conserving Small</u>	0.91	1.25
<u>OrbMol (Orb-v3 Conservative OMol)</u>	0.94	1.30
<u>MACE-ANI-CC</u>	1.04	1.30
<u>ωB97M-D3BJ/def2-QZVP</u>	1.18	1.59
<u>B3LYP-D3BJ/def2-QZVP</u>	1.41	1.84
<u>Egret-1e</u>	1.55	2.06

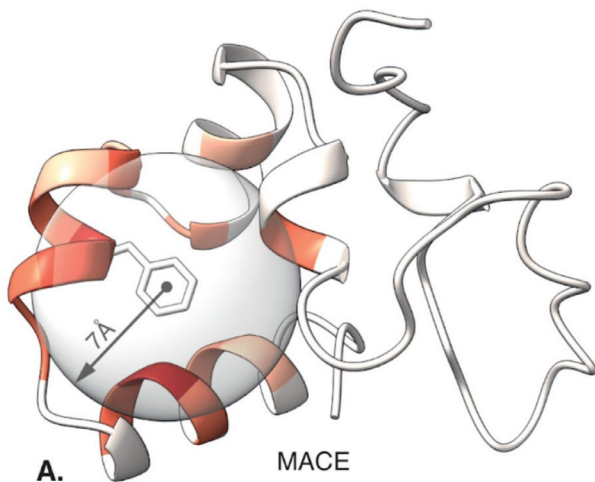
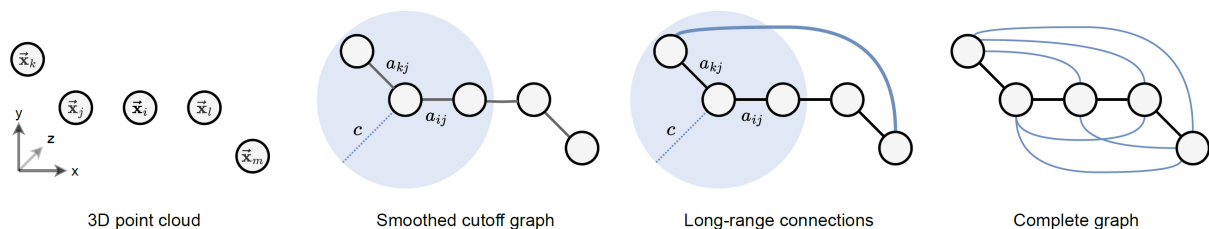
Limitations

- Poorer performance on ionization energies/electron affinity, spin-gap (i.e. charge and unpaired electrons)
- Short- and long-range NCIs
- Gas phase only

Long-range interactions: the next frontier

Most of the energy is accounted for by local environment

Recall cutoff criteria: *When should two nodes be connected by an edge?*



In eSEN, cutoff is 6 Å with 10 message passing layers:
theoretical “visibility” is 60 Å

But message-passing alone is unreliable

Option 1: Expand the cutoff

Maintain local nature of GNNs

GNNs become very slow at larger cutoffs:
hybrid approaches have been introduced

Option 2: Include explicit physics

Coulombic interactions are simple and quick to calculate,
given accurate calculations of atomic charges

Some reports where it does not improve models at all

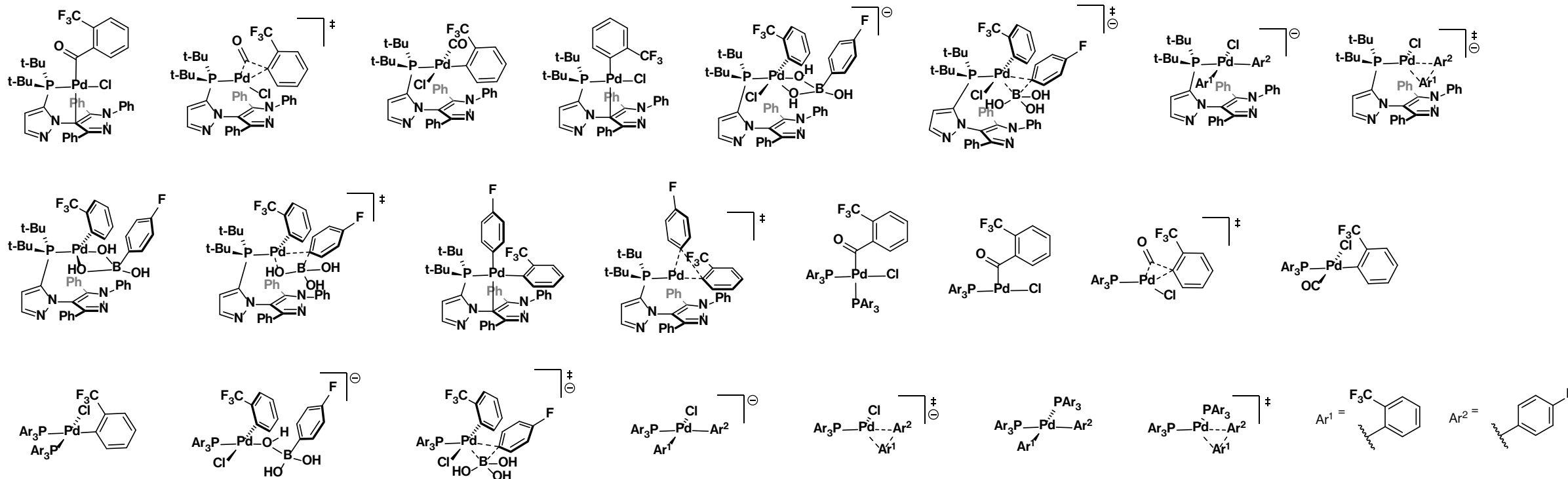
Option 3: Add a “nonlocal” features to each node

Add a learnable feature to account for non-local effects

GNN construction is still local, but each node gains an extra
learnable degree of flexibility

Various implementations exist

My experience with UMA: great for geometries, DFT high-level SP still needed



Workflow 1

B3LYP/6-311+G(2d,p)/SDD(Pd)/SMD(MeCN) //

B3LYP-D3BJ/6-31+G(d)/SDD(Pd)

Opt + Freq: 8.3 hours on 28 CPUs

Workflow 2

B3LYP/6-311+G(2d,p)/SDD(Pd)/SMD(MeCN) // **UMA-M**

Opt + Freq: 10 minutes on 1 GPU (cannot parallelize on GPUs)

MAD (kcal/mol)

GS structures: 0.48

TS structures: 0.89

Neutral: 0.61

Anionic: 0.79

Error is within chemical accuracy

Note: this is consistent with other benchmarks but not representative!

umadriver: an interface for routine calculations

 **umadriver** Public

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 **msh-yi** readme update

36f056d · 13 hours ago 3 Commits

old	initial commit	13 hours ago
umadriver	initial commit	13 hours ago
.gitignore	gitignore old dir	13 hours ago
README.md	readme update	13 hours ago
pyproject.toml	initial commit	13 hours ago
sample_manifest.yaml	readme update	13 hours ago

README

umadriver

Wrapper around **Meta's Universal Model for Atoms (UMA)** that drives an **ensemble** workflow per input XYZ file: geometry optimizations (incl. TS), single points, vibrational frequencies, quasi-RRHO thermochemistry, optional **Gaussian** input emission, and optional IRC tracing.

`umadriver` is exposed as a command-line tool and as a small Python API.

About

Wrapper for Meta's Universal Model for Atoms. Supports geometry opt, frequencies, thermochemistry

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